Assignment 2: Key Concepts and Problem Map

Chapter 2 — matrix algebra, inverses, rank, LU

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The problem map: 17 questions, six ideas

Concept cluster	Problems	The question it asks	Deciding test
A. Products & shapes	Q1 Q15 Q16	Is AB defined, and is it BA?	inner dimensions match; $AB \neq BA$ in general
B. Invertibility	Q2 Q3	does A^{-1} exist, and what is it?	$\det \neq 0$; 2×2 formula
C. Matrix equations	Q4 Q5 Q6	isolate X, respecting the side	$X = A^{-1}B$; order reverses for products
D. Rank & nullity	Q7 Q8 Q9	how many free variables?	Rank Theorem: $\text{rank} + \dim \text{Nul} = n$
E. LU factorization	Q10–Q14	factor $A = LU$, then use it	row additions recorded in L
F. Elementary matrices	Q17	row op as a matrix	perform the op on I

- Within a cluster the problems are the *same question with different arithmetic*: solve one carefully, the rest follow.

A · Products and shapes (Q1, Q15, Q16)

Q1 — compute AB, BA; then a T/F

$A = \left[\begin{smallmatrix} -1 & 1 \\ 4 & 1 \end{smallmatrix}\right]$, $B = \left[\begin{smallmatrix} 3 & 2 \\ 2 & -3 \end{smallmatrix}\right]$:

$$AB = \begin{bmatrix} -1\cdot 3+1\cdot 2 & -1\cdot 2+1\cdot(-3) \\ 4\cdot 3+1\cdot 2 & 4\cdot 2+1\cdot(-3) \end{bmatrix} = \left[\begin{smallmatrix} -1 & -5 \\ 14 & 5 \end{smallmatrix}\right], \quad BA = \left[\begin{smallmatrix} 5 & 5 \\ -14 & -1 \end{smallmatrix}\right] \Rightarrow AB \neq BA \Rightarrow \textbf{False}.$$

Q15 — one matrix, two products

$A = (-2, 3, 4)^\top$ (a column):
$A^\top A = (-2)^2 + 3^2 + 4^2 = \mathbf{29}$ (a 1×1 — enter a number); $AA^\top$ is the full 3×3 $[[4, -6, -8], [-6, 9, 12], [-8, 12, 16]]$.

Q16 — shapes decide

A, B: 4×9; C: 3×4. Defined: **A** $AB^\top$ $(4\times9)(9\times4)$, **D** $A^\top$ (9×4), **E** $B - A$ (same shape), **F** CA $(3\times4)(4\times9)$. Not defined: AC $(9 \neq 3)$, $C + A$ (shapes differ).

B · The 2×2 inverse test (Q2, Q3)

Q2 — does the inverse exist?

$M = \left[\begin{smallmatrix} 2 & 3 \\ 3 & 4 \end{smallmatrix}\right]$: $\det M = 2{\cdot}4 - 3{\cdot}3 = -1 \neq 0 \Rightarrow$ **exists**.

$$M^{-1} = \tfrac{1}{-1}\left[\begin{smallmatrix} 4 & -3 \\ -3 & 2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} -4 & 3 \\ 3 & -2 \end{smallmatrix}\right]$$

Check: $MM^{-1} = I$ ✓

Q3 — invertible for which c?

$A = \left[\begin{smallmatrix} 1 & 1 \\ c & c^2 \end{smallmatrix}\right]$:

$\det A = c^2 - c = c(c-1) \neq 0 \;\Rightarrow\; c \neq 0$ and $c \neq 1$.

One test, everywhere

$\det \neq 0$ is the Chapter 2 way to say "invertible" — the same fact the IMT states via pivots, rank, and trivial null space.

C · Matrix equations: isolate X without touching order (Q4, Q5, Q6)

Q5 — the warm-up

$AX = B$: left-multiply by A^{-1}, *keeping the order*: $X = A^{-1}B$ (never BA^{-1}).

Q4 — the real one

$AX(D + BX)^{-1} = C$: right-multiply by $(D + BX)$: $AX = C(D + BX) = CD + CBX$; gather X on the *right*: $(A - CB)X = CD$, so

$$X = (A - CB)^{-1}CD.$$

Q6 — a false shortcut

"$(A + B)^{-1} = A^{-1} + B^{-1}$": **False**. Counterexample (diagonals): $A = 2I$, $B = 3I$:

$$(A + B)^{-1} = \tfrac{1}{5}I \neq \tfrac{5}{6}I = A^{-1} + B^{-1}.$$

The two order rules

$(AB)^{-1} = B^{-1}A^{-1}$ reverses; and X factors out on the side it sits: $(A - CB)X = CD$, not $X(A - CB) = CD$.

D · The Rank Theorem, and Q8 step by step (Q7, Q8)

Rank Theorem: $\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n$ (number of columns)

- **Q7:** A is 10×9 with rank 7: nullity $= 9 - 7 = \mathbf{2}$.

Step 1 — RREF of the 4×4 matrix

$$R = \begin{bmatrix} 1 & 0 & 0 & 1/2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -13/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3 pivots; column 4 free $\Rightarrow$ nullity $= 1$.

Step 2 — substitute the free variable

$$\begin{aligned} x_1 + \tfrac{1}{2}x_4 = 0 &\quad \Rightarrow x_1 = -\tfrac{1}{2}x_4 \\ x_2 = 0 &\quad \Rightarrow x_2 = 0 \\ x_3 - \tfrac{13}{2}x_4 = 0 &\quad \Rightarrow x_3 = \tfrac{13}{2}x_4 \end{aligned}$$

Setting the free variable $x_4 = 1$: $\mathbf{v} = \left(-\tfrac{1}{2},\ 0,\ \tfrac{13}{2},\ 1\right)^\top$.

Step 3 — substitute $\mathbf{v}$ back into the ORIGINAL A (row by row)

$$\begin{aligned} &\text{row 1: } 2(-\tfrac{1}{2}) - 0 + 0 + 1 = 0 && \checkmark \\ &\text{row 2: } -\tfrac{5}{2} + \tfrac{13}{2} - 4 = 0 && \checkmark \\ &\text{row 3: } \tfrac{1}{2} - \tfrac{13}{2} + 6 = 0 && \checkmark \\ &\text{row 4: } -8(-\tfrac{1}{2}) - 0 - 2(\tfrac{13}{2}) + 9 = 0 && \checkmark \end{aligned}$$

all null vectors $= \operatorname{span}\{\mathbf{v}\}$.

D · Q8 and Q9: rank and nullity

The results (Q8's full derivation: previous slide):

	Result after elimination	rank	nullity $= n -$ rank
Q8 (4×4)	3 pivots; RREF has one zero row	3	$4 - 3 = \mathbf{1}$ (null vector $(-\frac{1}{2}, 0, \frac{13}{2}, 1)^\top$)
Q9 (6×4)	3 pivots; three zero rows	3	$4 - 3 = \mathbf{1}$ (null vector $(46, -5, -103, 37)^\top$)

- Both null vectors were verified: $A\mathbf{v} = \mathbf{0}$ exactly. Q9 is tall (6×4) — rank at most 4, and its rows are rank-3 dependent: the extra rows are combinations.

E · Q10 in detail: the two substitutions

Step 1 — $L\mathbf{y} = \mathbf{b}$ (forward)

$$\left[\begin{array}{cc|c} 1 & 0 & -13 \\ -1 & 1 & 9 \end{array}\right]$$

row 1: $y_1 = -13$

row 2: $-y_1 + y_2 = 9$

$\Rightarrow y_2 = 9 + y_1 = 9 - 13 = -4$

$\mathbf{y} = (-13, -4)^\top$

Step 2 — $U\mathbf{x} = \mathbf{y}$ (backward)

$$\left[\begin{array}{cc|c} 1 & 9 & -13 \\ 0 & 2 & -4 \end{array}\right]$$

row 2: $2x_2 = -4 \quad \Rightarrow x_2 = -2$

row 1: $x_1 + 9x_2 = -13 \quad \Rightarrow x_1 = -13 - 9(-2) = 5$

$\mathbf{x} = (5, -2)^\top$

Step 3 — substitute $\mathbf{x}$ back into the ORIGINAL $A = LU = \left[\begin{smallmatrix} 1 & 9 \\ -1 & -7 \end{smallmatrix}\right]$

row 1: $1 \cdot 5 + 9 \cdot (-2) = 5 - 18 = -13$ ✓

row 2: $-1 \cdot 5 + (-7)(-2) = -5 + 14 = 9$ ✓

$A\mathbf{x} = \mathbf{b}$ ✓

E · Solving with LU: two easy triangular steps (Q10)

Given $A = LU = \left[\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 9 \\ 0 & 2 \end{smallmatrix}\right]$, solve $A\mathbf{x} = \mathbf{b}$, $\mathbf{b} = (-13, 9)^\top$

Step 1 — $L\mathbf{y} = \mathbf{b}$ (forward): $y_1 = -13$; $-y_1 + y_2 = 9 \Rightarrow y_2 = 9 + (-13) = -4$.
Step 2 — $U\mathbf{x} = \mathbf{y}$ (backward): $2x_2 = -4 \Rightarrow x_2 = -2$;
$x_1 + 9x_2 = -13 \Rightarrow x_1 = -13 + 18 = 5$.

$$\mathbf{x} = (5, -2)^\top, \qquad \underbrace{\left[\begin{smallmatrix} 1 & 9 \\ -1 & -7 \end{smallmatrix}\right]}_{A=LU}\left[\begin{smallmatrix} 5 \\ -2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} -13 \\ 9 \end{smallmatrix}\right] \ \checkmark$$

- Two triangular sweeps replace one full elimination — that is the whole point of precomputing LU.

E · Building LU: record the multipliers (Q11)

Eliminate A (4×3) with row additions only; each multiplier is stored in L:

$$\begin{array}{llll} R_2 \leftarrow R_2 - 2R_1 & (L_{21} = 2) & R_3 \leftarrow R_3 - 3R_2 & (L_{32} = 3) \\ R_3 \leftarrow R_3 + 3R_1 & (L_{31} = -3) & R_4 \leftarrow R_4 + R_2 & (L_{42} = -1) \\ R_4 \leftarrow R_4 - 0R_1 & (L_{41} = 0) & R_4 \leftarrow R_4 - 2R_3 & (L_{43} = 2) \end{array}$$

$$U = \begin{bmatrix} 1 & -3 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -3 & 3 & 1 & 0 \\ 0 & -1 & 2 & 1 \end{bmatrix}, \qquad LU = A \checkmark$$

- U's last row is zero (no 4th pivot) — and L is 4×4 while U is 4×3: sizes differ (that is Q14 **False**).

E · Q11 reasoning: why L and U take this shape

Elimination = elementary matrices

Each addition $R_i \leftarrow R_i - mR_c$ is $E_i A$ with $E_i = I$ plus $-m$ at position (i, c). Stacking them:

$$E_4 E_3 E_2 E_1 A = U \implies A = \underbrace{(E_1^{-1} \cdots E_4^{-1})}_{=L} U.$$

Why L is unit lower triangular

E^{-1} of a row addition is the same addition with $+m$ at (i, c) — below the diagonal, ones on it. Products of such matrices stay unit lower triangular, and the multipliers m we stored during elimination appear unchanged in L.

Why U's last row is zero

A is 4×3: at most 3 pivots. Row 4 eliminated to $\mathbf{0}$ because it was already a combination of the others — L's row 4 holds the coefficients that rebuild A's row 4 from the *rows of* U:

$$\text{row}_4(A) = 0 \cdot \text{row}_1(U) + (-1) \cdot \text{row}_2(U) + 2 \cdot \text{row}_3(U)$$

$$= [0, -2, -2] \checkmark$$

After elimination only zeros remain in U's row 4, so U ends in a zero row and $\text{rank}(A) = 3$.

Why no swaps were needed

Every pivot was nonzero when reached (1, then 2, then -1). With a zero pivot we would need a row interchange $PA = LU$ instead.

E · Inverting via LU: reverse the order (Q12, Q13, Q14)

Q12 — $A^{-1} = U^{-1}L^{-1}$, one factor at a time

$L^{-1} = \left[\begin{smallmatrix} 1 & 0 \\ 6 & 1 \end{smallmatrix}\right]$ (flip sign of -6); $U^{-1} = \begin{bmatrix} -1/3 & -5/7 \\ 0 & 1/7 \end{bmatrix}$;

$$A^{-1} = U^{-1}L^{-1} = \left[\begin{smallmatrix} -1/3 & -5/7 \\ 0 & 1/7 \end{smallmatrix}\right] \left[\begin{smallmatrix} 1 & 0 \\ 6 & 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} -97/21 & -5/7 \\ 6/7 & 1/7 \end{smallmatrix}\right]$$

Check: $AA^{-1} = I$ ✓ (note: $L^{-1}U^{-1}$ would be wrong.)

Q13 — L diagonal

Pick A already in echelon form, $L = I_3$:
$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \\ 0 & 0 \end{bmatrix} = I_3 A$.

Q14 — same dimensions?

False: L is $m \times m$, U is $m \times n$ — Q11 gave 4×4 vs. 4×3.

F · A row operation as a matrix (Q17)

Build E by performing $R_4 + 3R_1 \to R_4$ on the identity

$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 3 & 0 & 0 & 1 \end{bmatrix}$ — the 3 sits in position $(4, 1)$: row 4 will collect $3\times$row 1.

Test: EA performs the operation

$A = \begin{bmatrix} 1 & -5 \\ 5 & -3 \\ 1 & -4 \\ 3 & -2 \end{bmatrix}$: rows 1–3 pass through, row 4 = row 4 + 3 row 1

$= [3, -2] + [3, -15] = [6, -17]$:

$$EA = \begin{bmatrix} 1 & -5 \\ 5 & -3 \\ 1 & -4 \\ 6 & -17 \end{bmatrix} \checkmark$$

- Left-multiplying by an elementary matrix = performing its row operation — the bridge between row reduction and matrix algebra, and the reason $[\,A \mid I\,] \sim [\,I \mid A^{-1}\,]$ works.

One table: the whole assignment

Idea	The one thing to remember
Products	inner dimensions match; $AB \neq BA$ (Q1); $A^\top A$ is 1×1 (Q15); shapes decide definedness (Q16)
Inverses	$\det \neq 0$ + the 2×2 formula (Q2, Q3); $(A+B)^{-1}$ is *not* $A^{-1}+B^{-1}$ (Q6)
Matrix equations	left-solve $AX=B$; for $AX(D+BX)^{-1}=C$: gather X on the right, $X=(A-CB)^{-1}CD$ (Q4, Q5)
Rank & nullity	$\text{rank}+\dim\text{Nul}=n$ (Q7: $9-7=2$; Q8, Q9: rank 3, nullity 1)
LU	additions build L, elimination builds U (Q11); solve in two triangular sweeps (Q10); $A^{-1}=U^{-1}L^{-1}$ reversed (Q12); sizes $m\times m$ vs $m\times n$ (Q13, Q14)
Elementary matrices	row op on I; EA performs it (Q17)

Review map — back to the class slides

Questions	Topic to review	Class slides (file · pages)
Q1 Q15 Q16	matrix operations, multiplication law, transposes	`CH2.1-2.2-2.3(1).pdf`: pp. 3–18 (esp. 9–15, 17–18)
Q2 Q3 Q6 Q14	2×2 inverse formula; inverses of products; elementary matrices	`CH2.1-2.2-2.3(1).pdf`: pp. 20–21, 23–26, 34–36
Q4 Q5	algebraic manipulation with inverses	`CH2.1-2.2-2.3(1).pdf`: pp. 23–25
Q17	elementary matrices as row operations	`CH2.1-2.2-2.3(1).pdf`: pp. 26–30
Q7 Q8 Q9	nullity, dimension, rank, Rank Theorem	`CH2.4-2.5-2.8-2.9.pdf`: pp. 29, 42–44
Q10–Q13	LU factorization and its uses	`CH2.4-2.5-2.8-2.9.pdf`: pp. 12–22

All files live in `RealSlidesUsed/course_files_export/`.

Answer key (instructor copy, reference only)

- Q1: $AB = [[-1, -5], [14, 5]]$; $BA = [[5, 5], [-14, -1]]$; False
- Q2: yes ($\det = -1$); $M^{-1} = [[-4, 3], [3, -2]]$
- Q3: $c \neq 0$ and $c \neq 1$
- Q4: $X = (A - CB)^{-1}CD$
- Q5: $X = A^{-1}B$
- Q6: False
- Q7: nullity $= 2$
- Q8: rank 3, nullity 1
- Q9: rank 3, nullity 1
- Q10: $\mathbf{x} = (5, -2)^{\top}$
- Q11: $L =$ $[[1, 0, 0, 0], [2, 1, 0, 0], [-3, 3, 1, 0], [0, -1, 2, 1]]$; $U = [[1, -3, 2], [0, 2, 0], [0, 0, -1], [0, 0, 0]]$
- Q12: $A^{-1} = [[-97/21, -5/7], [6/7, 1/7]]$
- Q13: e.g. $A = [[1, 2], [0, 3], [0, 0]]$
- Q14: False
- Q15: $A^{\top}A = 29$; $AA^{\top} = [[4, -6, -8], [-6, 9, 12], [-8, 12, 16]]$
- Q16: A, D, E, F
- Q17: $E = [[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 1, 0], [3, 0, 0, 1]]$; $EA = [[1, -5], [5, -3], [1, -4], [6, -17]]$

All answers verified by exact-arithmetic scripts (`chapter2_solutions/`).

Thank you!

Questions? Comments? Corrections?

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